

A Grammar-Driven Rule-Based Stemmer for Pashto

Khairullah Ibrahim Khail

Institute of Management Sciences, Peshawar, Pakistan

ibrahimkhil976@gmail.com

ABSTRACT

Pashto has roughly forty million speakers and very few text-processing tools. The only stemmer published for the language applies nine affix rules without a lexicon and without any means of choosing between competing analyses. We present a rule-based Pashto stemmer built from an affix inventory of 117 rules drawn from published grammars, in which every rule records its source, its productivity, and whether it may be applied at all. Candidate stems are generated separately from the decision of which candidate to keep, so a shallower analysis can displace a deeper one. No machine learning is used.

Since no annotated set for Pashto stemming existed, we built two. A development set of 2,912 word types and a held-out set of 500 types sampled from news text were each annotated independently by three native-speaker Pashto educators, who agreed on 96.88% of stems ($\kappa = 0.969$). The held-out set was annotated with its reference column empty and was not consulted while the system was being developed. 499 of its 500 types fall within the stemming task and are the denominator below; the remaining one needs a different word and is reported apart.

On that set the proposed system reaches 71.34% exact-match accuracy against 40.48% for a faithful reimplementation of the prior system, and 70.6% against 16.9% on the types that carry an affix. Paice's under-stemming index falls from 0.889 to 0.556.

A secondary finding concerns evaluation itself. The protocol used in the prior work — presenting system output to native speakers and asking whether it is acceptable — assigns a score of 100% to a system that modifies no word at all. We report results under both that protocol and exact match against a reference fixed in advance, and recommend the latter for Pashto stemming.

Keywords: Pashto, stemming, morphological analysis, rule-based methods, low-resource languages, evaluation methodology, information retrieval

1. INTRODUCTION

A reader searching for documents about houses does not distinguish between house, houses and of the houses. A retrieval system does, because to the system these are three unrelated strings. Stemming removes that difference by stripping the endings a word carries, so that related forms collapse onto a single index term. For English the problem has been settled since Porter's algorithm [2], which remains in production use after four decades.

For Pashto almost nothing exists. The language is spoken by approximately forty million people across Afghanistan and Pakistan, is one of the two official languages of Afghanistan, and has a growing volume of digital text in news and social media. It also has a rich inflectional system, which is precisely the condition under which stemming matters. A Pashto retrieval system without stemming will miss most relevant documents; a classifier will treat the singular and plural of a key term as

unrelated features and learn each from less data than it has.

The single published Pashto stemmer, by Aslamzai and Saad [1], applies nine affix rules in a fixed order with no lexicon and no selection stage. Our reimplementation of those rules leaves most words untouched, and damages a substantial fraction of the words it does modify.

Three properties of Pashto make a naive approach fail.

A. Five letters for one sound

Pashto writes five distinct characters conventionally called *ye*. They are not interchangeable; they carry gender, number and case:

TABLE I. THE FIVE PASHTO YE LETTERS

Letter	Example	Function
ی	سری	masculine singular direct
ي	سري	masculine plural/oblique; 3rd person present
ې	ښکلې	feminine; plural
ۍ	نجلۍ	feminine singular, one noun class
ئ	ووايئ	2nd person plural imperative

Normalization pipelines written for Persian or Urdu routinely merge these into a single character. For Pashto that destroys the information a stemmer needs: *سری* (man) and *سري* (men) become indistinguishable. Our system preserves all five.

B. Endings that are not endings

The same final letter may be an affix in one word and part of the root in another, with nothing in the surface form to separate the two cases: *لنډه* reduces to *لنډ*, where *ه* marks agreement, while *سيمه* must remain *سيمه*, because there the *ه* belongs to the word. A native speaker resolves this by knowing the word; a rule conditioned on the surface form cannot.

C. Verbs whose stems cannot be derived

Pashto distinguishes weak verbs, whose stems follow from the infinitive, from strong verbs, whose present stem must be known: *ليدل* has the present stem *وين*, and *تلل* has *خ*. No affix rule derives *وين* from *ليدل*. These verbs are frequent and must be listed.

D. Contributions

- An affix inventory of 117 rules in which each entry records its grammatical source, its productivity, a confidence level, and an explicit decision about whether it may be stripped. Twenty-five entries are documented but never applied.
- Two annotated evaluation sets for Pashto stemming, one of which was annotated blind and never used during development.
- An evaluation protocol that separates performance on word types carrying an affix from performance on types that do not, because overall accuracy alone rewards inactivity in a language where nearly half of word types are uninflected.
- A quantitative account of what each component contributes, including two components that we retain

despite a measured cost and one rule that we implemented, measured and removed.

- The observation that the evaluation protocol used in prior Pashto stemming work cannot distinguish a working stemmer from an inert one.

2. RELATED WORK

A. Rule-based stemming

Lovins [3] published the first English stemmer, a single-pass longest-match algorithm over 294 endings. Porter [2] replaced it with a five-step procedure whose conditions depend on a measure of syllable count. Its durability is instructive: the algorithm is small, fast, entirely inspectable, and makes no claim to produce real words. Porter maps both relational and relate to relat, which is not an English word. The output of a stemmer is a class label rather than a lemma, and confusing the two is a recurring error in work on morphologically rich languages.

Paice [4] contributed both an algorithm and, more consequentially for the present work, an evaluation method [5]. His under-stemming index counts related word pairs a system fails to conflate, and his over-stemming index counts unrelated pairs it wrongly merges. Because the two move in opposite directions, reporting both is more informative than accuracy alone.

B. Arabic-script languages

Khoja and Garside [6] built an Arabic root extractor that strips affixes and matches the remainder against trilateral patterns. Larkey, Ballesteros and Connell [7] showed that light stemming, removing only a short list of prefixes and suffixes, outperforms aggressive root extraction for retrieval. Their conclusion shaped the design adopted here: removing less, but removing it reliably, is preferable to removing more. For Persian, Sharifloo and Shamsfard [8] used a bottom-up parser over a morpheme lexicon. The pattern common to these systems is that a lexicon resolves ambiguity that rules alone cannot.

C. Pashto

Work on Pashto stemming is confined, to our knowledge, to Aslamzai and Saad [1]. Because that system is the baseline for everything reported here, we describe it in full.

Their algorithm applies nine rules, four removing suffixes and five removing prefixes, each with a length condition. Rule 1 removes one of eleven suffixes from words of five or more characters; Rule 2 replaces a final *ي* with *ه* in words of four or more; Rule 3 removes *من*; Rule 4 removes *يز* and appends *ه*; Rules 5, 6, 7 and 9 remove prefixes; and Rule 8 reduces a reduplicated word to one copy.

Reproducing the baseline exactly mattered, since a comparison against a misimplementation would be worthless. Our implementation reproduces all eight worked examples given in the paper. Four points in the publication required a decision, and we record them so that they can be checked:

- The order of prefix and suffix removal is not stated. We remove suffixes first, because the paper's own example *موتروان* → *موتر* only succeeds in that order; with prefixes first, the Rule 5 prefix *م* removes the first letter of *موتر*.
- Rule 3 is stated for words of “more than 5 characters” but its example *واکمن* has exactly five, so we implement the condition as five or more.

- The Rule 5 prefix list contains a bare *م*, which contradicts the *موتروان* example above. We exclude it, giving the baseline the reading its own example implies rather than the one that would lower its score.
- Rule 4 instructs the reader to remove *يز* and append *ه*. Its first example follows this, but its second gives *ه. دوديز* → *دود* with no *ه*.

Rule 2 is also linguistically incorrect as published. It maps the masculine plural adjective *بنکله* to *بنکلي*, which is the feminine singular; Tegey and Robson [9] give the masculine singular *بنکلی* as the base form of this class. The rule is corrected in our inventory.

Finally, the reported accuracy of 87% is not comparable with the figures in this paper, and the difference lies in the protocol rather than in the algorithm. We return to this in Section VI.

D. Grammatical sources

The affix inventory was assembled from published descriptions rather than by inspecting a corpus. Table II lists the sources used and the tag each carries in the released inventory file.

Of the 117 rules, 39 cite one of those published descriptions directly. A further 30 reference a section of this project's own affix-inventory document, which collates the same literature but is not itself a publication, and 49 carry no reference at all. We separate the three cases rather than report one citation figure, because only the first is a citation in the ordinary sense.

TABLE II. SOURCES FOR THE AFFIX INVENTORY

Source	Tag
Tegey & Robson, A Reference Grammar of Pashto [9]	[T&R]
Robson & Tegey, “Pashto” [10]	[R&T]
David, Descriptive Grammar of Pashto [11]	—
Naeem & Khan, derivational morphology [12]	[N&K]
Khan, Pashto diminutives [13]	[Khan23]
Wiktionary / kaikki.org Pashto suffixes	[W]
Apertium apertium-pus	[AP]

3. PASHTO MORPHOLOGY

This section covers the parts of the morphology that bear on the design. It is not a complete grammatical description.

A. Orthography

Beyond the Arabic inventory, Pashto adds retroflex and other consonants that occur in neither Arabic nor Persian: *ت د ږ ځ ښ څ ښ ک*. A word containing any of these is certainly Pashto rather than a loan, and the system uses that fact.

A complication became apparent only late in this work. The same suffix is written with different ye letters in different registers: dictionary orthography prefers لومري where news orthography writes لومري. A rule spelled in one convention does not fire on the other. The system therefore compares a folded key when deciding whether a rule applies, while the letters it writes out are never altered. Section VII quantifies the effect.

B. Nominal inflection

Tegey and Robson divide Pashto nouns into masculine classes M1–M4 and feminine classes F1–F3, each with its own plural and oblique endings. The endings the system removes most often are ونه and ونو (masculine plural and its oblique), انو and ان (animate plural and oblique), ې (feminine plural) and و (oblique plural). Nominal inflection is not a simple suffix-substitution system: gender, number, noun class and the final shape of the stem can all trigger stem alternation.

C. Verbal morphology

Two productive patterns form verbs from nouns and adjectives. The بدل intransitive and ول transitive patterns are regular, so both are handled by rule and neither appears in the dictionary: جور بدل → جور and جورول → جورول. For strong verbs the present stem cannot be predicted, and these are listed: کول / ځ / تلل / وين / ليدل / غوار / غوښتل / کو.

D. Loanwords

Pashto has borrowed heavily from Arabic, Persian and English, and loanwords are the principal remaining source of error, because their endings frequently resemble Pashto affixes. The correct stem of منشي is منشي, not منش; of کورس is کورس, not کور, which is a different word meaning house; and of بېجينگ (Beijing) is the word itself, not جينگ. Arabic broken plurals are a related problem, formed by changing the internal vowel pattern rather than by adding material, so that no suffix rule can reach them.

4. METHOD

Fig. 1 shows the order in which a word passes through the system. Three of the stages can return a stem on their own, and the remaining word goes to the generate-and-select

stages that do the general work. Every stage can be disabled independently, which is what makes the component study in Section VII possible.

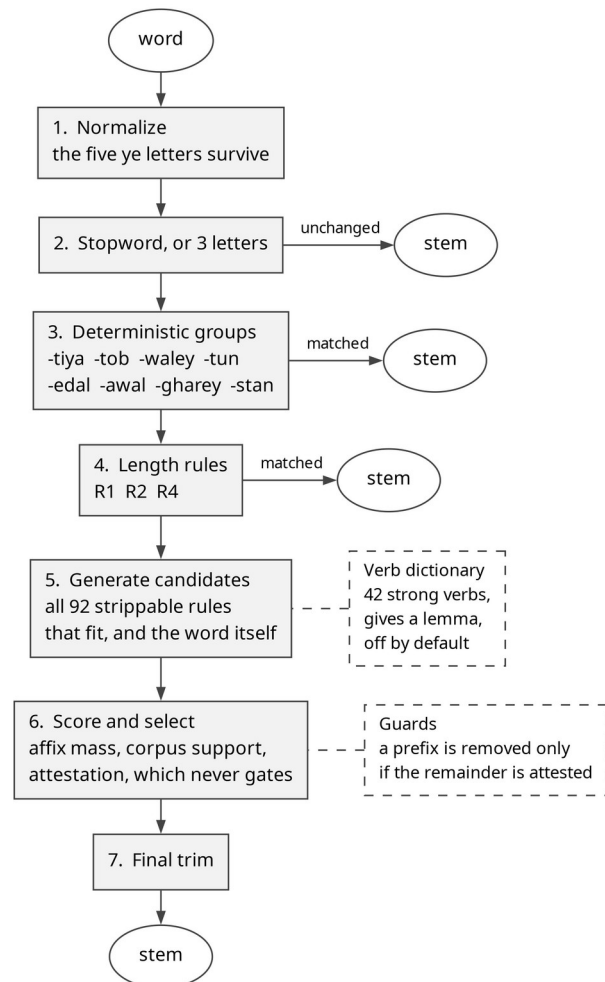


Fig. 1. The pipeline. Stages 2–4 can each return a stem and stop; everything else reaches stage 5, where all applicable rules are applied and the results are then scored against one another. The two dashed boxes are not stages: the guards constrain which strips the earlier stages may make, and the irregular-verb dictionary is an optional lemmatization mode that is off by default. Affixes are transliterated in the figure for legibility and given in Pashto script in the text.

A. Normalization

Normalization removes differences that carry no information: Arabic diacritics, the tatweel elongation character, zero-width joiners, and the several Unicode encodings of the same letter. The Arabic kaf is mapped to the Pashto keheh, the alef-hamza forms to bare alef, and the Urdu letters common in Pashto written in Pakistan (ٹ

ڙ ځ and the Persian گ) to their Pashto counterparts. These are encoding corrections rather than changes of spelling.

What normalization does not do is merge the ye letters. We verified that no ye character is altered in any cell of either evaluation set. The aggressive fold remains available behind a configuration flag so that the cost of the error reported in the literature can be measured rather than asserted.

B. The affix inventory

The inventory holds 107 suffix rules and 10 prefix rules. Each rule records the affix and the side it attaches to; whether it is inflectional or derivational; a minimum surviving stem length; the part of speech the affix signals, where known; its productivity (productive, restricted, lexicalized or borrowed); a confidence level; and whether it may be stripped at all.

Table III gives the schema in full, with three entries written out. The last field does real work. Twenty-five entries are documented but never applied, because applying them was measured to be harmful or because the grammatical source provides no attested example. The lexicalized diminutive وکی, the last column of Table III, is

one: it survives only in fixed words such as وړوکی, so removing it by rule damages them. Recording the affix and refusing to strip it is not the same as omitting it, because the inventory is also a description of the language. The minimum stem length does equally real work: without it the verb کېدل reduces to the single letter ک.

TABLE III. THE RULE SCHEMA, WITH THREE ENTRIES WRITTEN OUT

Field	ونه	توب	وکی
side	suffix	suffix	suffix
category	inflectional	derivational	derivational
min. stem	2	3	3
POS signalled	N	N	N
productivity	productive	productive	lexicalized
confidence	high	high	low
strip allowed	yes	yes	no
source	[T&R] M2	[N&K]	inventory §4
example	کورونه → کور	مینتوب → مین	وروکی unchanged

The third entry is documented and never applied. The header row and the last two rows are in Pashto script.

Table IV groups the 117 rules by what they do. Derivation accounts for 71 of them against 37 inflectional rules, which is the opposite of what a stemmer for English would need and follows from where Pashto puts its productive morphology. The grouping also shows where

the gated entries are concentrated: 16 of the 25 are noun-forming derivational suffixes, which is the area where lexicalization is commonest and where a rule is most likely to be documented in a grammar and still be unsafe to apply.

TABLE IV. THE 117 RULES BY FUNCTION

Group	Rules	Published	Kept	Examples
Nominal inflection	23	15	0	ونه انوې
Verbal inflection	13	7	1	بدل ول
Derivation: nouns	56	14	16	توب تیا تون
Derivation: adjectives	14	2	4	یز مند ناک
Derivation: verbs	1	1	0	لې
Prefixes	10	0	4	هم بیا نیم
Total	117	39	25	

“Published” counts the rules citing a published grammar or lexical resource; “kept” counts those documented and never stripped. The full inventory is Appendix A.

C. Length rules

Three rules were contributed by the author, a native speaker of Pashto, from his own judgement, and then tested against the annotated data. A word of three letters or fewer is returned unchanged. A word of four or five letters ending in one of the five *ye* letters or in *ه* loses that letter.

A word ending in *ستان* loses that suffix, and stripping stops there.

The last condition is necessary: allowing the general rules to continue produced *ترک* → *ترکمښتان* rather than *ترکمښتن*. The second rule originally included *ل*, which we removed after measuring it. On a verb *ل* is the infinitive ending, but on a noun it is part of the root, and the rule could not distinguish the two: it reduced *کابل* to *کاب* and *لام* to *لامل*, accounting for 51 over-strips on its own.

A fourth rule was proposed and rejected. It would have removed the last three characters of any non-Pashto word of seven characters or more. Measured against three independent references, removing a documented affix instead was more accurate in both careful annotations (45.0% against 25.0% on the development set, 51.6% against 43.5% on the held-out set). Every word the rule damaged carried a genuine affix: *مسوولیت* became *مسوو* where the affix is *یت*, and *اعلانات* became *اعلا* where the affix is *ات*. The rule fired before the inventory was consulted and cut through the root.

D. Candidate generation and selection

Generation and selection are separate, which is the central design decision borrowed from the Persian and Urdu hybrid systems. The generator applies every rule that fits and returns all distinct forms reached, including the original word with nothing removed. For *کورونو* the pool contains the untouched form, the result of removing *ونو*, and the result of removing *و* alone. Nothing is decided at this stage, which is what allows a shallower analysis to displace a deeper one.

Selection scores each candidate on how much affix material was removed, whether other words in the corpus are built on the same stem, and whether the stem itself is attested. Two corrections to this stage were necessary during development and are worth recording, since both are easy to get wrong.

First, an early version required the stem to be an attested Pashto word before accepting a strip. That is incorrect: the output of a stemmer is a class label, and Porter's *relat* is not an English word either. Attestation now raises a candidate's score without gating it, and the system freely produces stems that are not words, including *پوهن* and *لوبغار*, *ښوون*.

Second, the unstripped input word is almost always present in the corpus, so awarding it the full attestation bonus made inaction the default winner and concealed every single-letter inflection. Presence in a corpus is evidence that a string is a word, not that it is a stem, and the no-strip candidate now receives a much smaller bonus.

A small number of affix groups bypass the scorer entirely. The abstract-noun and place suffixes *تیا*, *توب*, *تون* are long and unambiguous, so no competing analysis exists for the scorer to weigh; leaving them arbitrated meant that *پوهنتون* and *روغتیا* remained unchanged because *پوهن* and *روغ* are absent from the lexicon.

E. Guards

Two guards were added after error analysis. A prefix is removed only when what remains is itself an attested word. Without it the system strips *لا* from *لانسیت*, leaving *نسیت*, and from *لانتها*, leaving *انتها*; with it, both words are returned whole. The guard is conservative in both directions, since *لانتها* genuinely is *لا* + *انتها*, and we accept the missed strip in exchange for not cutting into loanwords.

The second guard concerns the interaction between the prefix rules and the length rules. A word of four or five letters that begins with a removable prefix is passed to the full inventory instead of simply losing its last letter. Only a prefix that may actually be removed counts: *نا* is kept rather than stripped, so *ناوره* is still handled by the length rule and becomes *ناور*, which is what the annotation gives.

A third component was implemented and then removed. A compound splitter returned the head of a word it judged to be a compound, which conflicted with the annotation policy's decision to keep compounds whole. Measured, it fired on seven types across the two evaluation sets and was wrong on all seven, usually because the “head” it returned was itself a suffix: *نړیوال*.

→ وال ښووليم ښووليم. Disabling it is worth 0.22 points on the development set and 0.20 on held-out text. Splitting Pashto compounds reliably needs a lexicon of free forms that we do not have, and guessing at the boundary was worse than leaving it alone.

F. The verb dictionary and the scope of the task

The system includes a dictionary of 42 strong verbs, of which 34 are checked against a printed grammar and 8 are marked unverified. The dictionary maps an inflected form to its verb: ښوولې → ښوول. That output is a lemma, and substituting one word for another is lemmatization rather than stemming. On the word types whose reference is a truncation, the dictionary decides the output on 61 of them and is right on 11, because it supplies the wrong kind of answer; it costs 1.03 points on the development set and 1.00 on held-out text.

The dictionary is therefore disabled by default and provided as an explicit lemmatization mode. We report its contribution separately in Section VII rather than folding it into the stemming result.

5. DATA AND ANNOTATION

No annotated set for Pashto stemming was available, so two were constructed. The distinction between them determines how the results should be read.

A. Corpus

Word frequencies were taken from a corpus of Pashto news and general text assembled from VOA Pashto, BBC Pashto and a topic-labelled collection, totalling approximately 5.4 million tokens and 68,850 distinct word types. The lexicon shipped with the system is a 5,443-type frequency list drawn from this material.

Building a larger lexicon from the full corpus was attempted and made results worse. The reason is instructive: a raw corpus contains inflected forms as well as base forms, so with 48,000 types the attestation signal ceased to mean “this is a stem” and began rewarding over-stripped forms that happened to occur. The smaller list was better by being sparser.

B. Development set

2,912 word types were annotated independently by three native speakers of Pashto before the rule inventory was settled: the author, the principal of a Pashto-medium school, and a university teacher of the language. None of the three saw another's work, and none saw any system output. The author then adjudicated the disagreements and produced the released reference.

Two of the three passes are released as separate files, and the agreement between them can therefore be recomputed rather than asserted. On the 2,725 types both annotated, they chose the same stem for 96.88%, which is a Cohen's κ [14] of 0.969. Reduced to the decision that matters most — whether the word carries an affix at all — they agree on 98.24%, $\kappa = 0.965$. The two coefficients answer different questions. Chance agreement on the full stem string is 0.001, because the label space is every substring of the word, so there κ adds little to the raw figure; on the binary decision chance agreement is 0.503, and the κ is doing real work. Both fall in the band Landis and Koch [15] describe as almost perfect agreement.

The adjudicated reference stays close to both: 93.64% identical to the school principal's pass and 94.98% to the university teacher's. Where it departs from both, it follows the written annotation policy, and the 42 policy decisions taken during the project are logged individually with their reasons.

One part of the construction is circular and we state it rather than leave it to be found. 116 verb forms were revised for consistency with the verb analysis, and 81 of those were changed to the reading the system itself produces, which is 3.0% of the stemming types. That is a real dependency between the reference and the system, it is confined to verbs, and it is why the held-out set rather than this one is treated as the result.

C. Held-out set

500 word types were sampled at random from the news corpus under three constraints: a frequency of at least five, to exclude typographical noise; a length of at least three characters; and no overlap with the development set. The sample was annotated by the same three native speakers, working independently and with the reference field initially empty, so that nothing anchored them to any system's output. They agreed on it, and the author adjudicated the remainder as before. The set was not consulted during development, and every figure reported for it comes from a single run of the finished system. The released file carries two reference columns, `gold_stem` and `expert_stem`, whose contents are identical: they record the one adjudicated reference under two names.

The first pass was audited by comparing every removal against the affix inventory. On the released first-pass file, 50 of 299 removals, 16.7%, match no documented affix: words such as مزاج and حارث, which carry no affix at all, had been shortened. Thirty rows were revised in response, and the figure falls to 33 of 287, or 11.5%. The same audit on the development set leaves 105 of 1,355 removals, or 7.7%, unaccounted for.

That audit procedure is reusable and we recommend it. For each annotated pair, the material removed is compared against the affix inventory; a removal matching no affix is either an annotation error or evidence that the inventory is incomplete, and inspection separates the two. Applied to the development set, the same procedure identified six affixes the inventory lacked.

The 33 removals that remain unaccounted for in the final held-out set are a live defect, not a resolved one. Each is either an annotation error, in which case the reported accuracy is measured against a reference that is wrong on that row, or a gap in the inventory, in which case the inventory is incomplete. They are released as a list so that the question is open to inspection rather than settled by assertion.

D. Annotation policy

The policy the annotations follow is released with the data. Its main provisions are that the output is a stem and need not be a word; that Arabic broken plurals are handled by dictionary lookup rather than by rule; that proper nouns receive no special treatment, so *افغانستان* loses its *ستان* like any other word; that compounds are kept whole; and that where two analyses are both defensible, the shallower is preferred.

6. EVALUATION PROTOCOL

A. Why exact-match accuracy alone is insufficient

The natural measure is the proportion of word types for which the system produces the annotated stem. On its own it is misleading. Nearly half of Pashto word types carry no affix and should be returned unchanged, so a system that modifies nothing scores 47.77% on the development set and 37.27% on the held-out set. Every letter a system declines to remove earns it credit.

Tuning against overall accuracy therefore drives a stemmer towards inactivity. During this work an apparent improvement of nearly three points proved, on inspection, to be exactly that: the system had become more

conservative, losing seven points on the word types requiring a strip while gaining twelve on those requiring none.

All results below report three figures: accuracy on the types the annotation strips, accuracy on the types it leaves unchanged, and overall accuracy. The first cannot be obtained by inaction.

B. The scope of the stemming task

Stemming removes characters; it cannot insert them. Where the reference is an Arabic broken plural or a suppletive form — *نجلۍ* → *نجونو*, *اثر* → *اټارو* — the answer is a lemma that no affix rule can produce. Marking a truncating stemmer wrong on those types penalises it for not being a lemmatizer.

We therefore score the two tasks separately. A type belongs to the stemming task when the reference can be obtained by removing characters from the word, comparison being insensitive to which letter is written. By this criterion the development set contributes 2,717 stemming types and 195 lemma types, and the held-out set 499 and 1. Results for the lemma types are reported separately in Section VII.

C. Two protocols compared

Aslamzai and Saad report 87% accuracy; our reimplement of the same nine rules reaches 40.48% on held-out data. The difference is entirely methodological, and the point is important enough to measure rather than assert.

Protocol A, used throughout this paper, fixes the reference before any system output is seen and counts a result only if it matches exactly. Protocol B reproduces theirs: a judge is shown the output and decides whether it is an acceptable analysis. We implement Protocol B reproducibly by counting an output as acceptable when nothing was removed that is not a documented affix, and by treating an unchanged word as always acceptable, which is how a judge shown an unchanged word would score it.

TABLE V. THE SAME 499 HELD-OUT WORD TYPES UNDER BOTH PROTOCOLS

System	A: exact	B: judged	Modified
Modifies nothing	37.27%	100.00%	0.0%
Aslamzai & Saad [1]	40.48%	80.76%	37.5%
Proposed	71.34%	98.60%	68.3%

The final column gives the proportion of types the system modifies at all.

The first row of Table V is the finding. Under Protocol B a system that modifies no word scores 100%, because every unchanged word is acceptable to a judge and it produces

nothing else. The protocol cannot separate a working stemmer from an inert one, and a figure obtained under it cannot be compared with an exact-match figure.

Nothing here shows the published 87% to be incorrect. It shows that the measurement answers a different question. We note that the proposed system also outperforms the baseline under Protocol B, by 98.60% to 80.76%, so the conclusion does not depend on the choice.

7. RESULTS

A. Held-out evaluation

Table VI reports the held-out set, which is the primary result: 499 word types from news text, annotated with the reference field initially empty and never used during development.

TABLE VI. HELD-OUT EVALUATION, 499 WORD TYPES

System	Accuracy	Affixed	Bare	UI
Modifies nothing	37.27%	0.0%	100.0%	1.000
Aslamzai & Saad	40.48%	16.9%	80.1%	0.889
Proposed	71.34%	70.6%	72.6%	0.556

“Affixed” and “Bare” are accuracy on the types the annotation changes and leaves unchanged.

The proposed system improves on the baseline by 30.9 points overall. The separation of the two halves is more informative than the overall figure: on the types that require a strip the baseline reaches 16.9% and the

proposed system 70.6%, a factor of four. Paice’s understemming index falls from 0.889 to 0.556, and overstemming remains below 4×10^{-5} .

B. Development set

TABLE VII. DEVELOPMENT SET, 2,717 STEMMING TYPES

System	Accuracy	Affixed	Bare	UI
Modifies nothing	47.77%	0.0%	100.0%	1.000
Aslamzai & Saad	50.68%	12.2%	92.8%	0.919
Proposed	79.43%	77.0%	82.0%	0.314

Table VII reports the development set. The figure is higher than the held-out result. The revisions described in Section V.B account for 0.07 points of that gap, so the explanation lies elsewhere: it is orthographic, and Section VII.D gives it. The development set is reported for

comparability with the baseline, which is scored identically, and because Paice’s indices are more stable on the larger set. On the types requiring a strip the margin is wider still: 77.0% against 12.2%, a factor of six.

C. Component study

TABLE VIII. EACH COMPONENT REMOVED IN TURN

Configuration	Development	Held-out
Full system	79.43%	71.34%
– suffix rules	71.99% (–7.43)	58.92% (–12.42)
– length rules	76.48% (–2.94)	69.34% (–2.00)
– corpus lexicon	78.32% (–1.10)	71.74% (+0.40)
– uniform-strip group	79.13% (–0.29)	71.14% (–0.20)
– prefix rules	79.39% (–0.04)	71.54% (+0.20)
+ verb dictionary	78.40% (–1.03)	70.34% (–1.00)

Changes in percentage points.

The suffix inventory does most of the work, and removing it costs 12.4 points on held-out text. The author’s length rules contribute between two and three points on both sets.

Three rows record decisions taken against the measured score, and we state them explicitly rather than quietly adopting whichever setting scored best.

Prefix stripping costs a few hundredths of a point on the development set. We retain it because هم, بيا, نيم and

لا are genuine Pashto prefixes, and removing the rules would amount to claiming the language has none; the attestation guard described in Section IV.E is what makes them safe. The small effect is itself explained by the inventory: of the ten prefix rules, four are documented and never stripped. نا, بې, غير and نه are negation prefixes that the annotation policy keeps, because ناقانونه is not an instance

of قانون but its opposite, and سر is a free noun as often as it is a prefix.

The corpus lexicon is worth 1.10 points on the development set and -0.40 on held-out text. Its contribution is now marginal, because the uniform-strip and length rules decide most cases before the scorer is consulted. We report this plainly: the system should not be described as lexicon-driven.

Enabling the verb dictionary costs about one point on both sets, for the reason given in Section IV.F. Its proper contribution is to the lemma types, where it answers 51 of the 195 in the development set against 8 without it.

D. Orthography

The gap between the two evaluation sets has one dominant cause. Of the 500 held-out news types, 118 contain the hard ye يي; of the 2,912 development types, none do. The inventory contained thirteen rules ending in the narrow ye ی and none ending in يي, so a large class of news words passed through untouched: لومري and تاريخي were returned unchanged.

Making rule matching insensitive to which ye is written, while leaving the output letters untouched, corrected this. On the development set, which is written entirely in one convention, exact matching scores 0.40 points higher; on held-out news text the folded comparison is worth a point, so it earns its place on the set that reflects real text. This is the clearest instance in the project of a defect that a single-source evaluation set could not have revealed.

8. DISCUSSION

A. Correct rules and under-determined rules

A per-rule analysis separates two kinds of failure, and the distinction matters for how the remaining error should be addressed.

Some rules are simply wrong. Three were identified and removed during this work: a standalone rule for the verbal ل, which only ever applied to nouns where ل belongs to the root and so turned كابل into كاب; a rule removing نى as a unit, where the affix is usually the ی alone; and a rule restoring a feminine ه to any stem that had lost one, which produced كورونه → كورونو. These were defects rather than ambiguity. We give the words rather than firing counts because the counts were

measured against an inventory that has since changed and cannot be recomputed from the released code.

Other rules are correct but under-determined. The rule removing a final ه from a short word fires 131 times on the development set and is correct 86 times. لنه → لنه is right and خواره → خوار is wrong, and the two words are indistinguishable in shape. Removing the rule loses more than it saves: overall accuracy falls by 1.62 points. No reformulation conditioned on the surface form can help; only a lexical resource can.

B. What limits the current accuracy

The 559 development-set errors divide into three kinds. 317 of them, 57%, are over-strips, where the system removed more than the reference does; 201, 36%, are under-strips, and 101 of those are words the system left untouched altogether; the remaining 41 are the same length as the reference but differ in which letters survive. The held-out set behaves the same way, 52% and 43%. Over-stripping is therefore the larger problem on both sets, which is what one would expect of a system whose scorer rewards removing affix material, and it is concentrated on the يي/ی ambiguity and on loanwords rather than on missing affixes.

We tested the obvious remedy and report the result because it is negative. A 257-entry list of development-set words that should be left unchanged raises development accuracy by 8.5 points and held-out accuracy by zero: of the 51 held-out types needing the same treatment, none appear in the list. The list is pure memorisation of one sample. The generalizing form of the same idea is a named-entity and loanword lexicon, since the words concerned are overwhelmingly names and borrowings, and we leave that to future work.

9. LIMITATIONS

The evaluation is on word types rather than running text, so no measurement of retrieval effectiveness is reported, and the relationship between the intrinsic gains shown here and downstream performance remains untested.

Inter-annotator agreement is measured on the development set only. The annotators' individual passes on the held-out set were not retained, so no agreement figure is available for the set the main result rests on.

Of 117 affix rules, 39 cite a published description directly and 49 carry no reference at all. Of 42 verbs in the dictionary, 34 are checked against a printed grammar and 8 are marked unverified in the released documentation.

We prefer to label these rather than to present them as settled.

Compounds are kept whole, which is a policy choice rather than a solution, and the annotation does not follow it perfectly: an audit during development found types where the reference itself splits a compound. The released audit script reports every removal that matches no documented affix, which is where such cases surface.

The corpus is drawn from news and general text, so the vocabulary of specialised domains is under-represented.

10. CONCLUSION

We presented a rule-based stemmer for Pashto built from an affix inventory of 117 rules taken from published grammars, each recording its source, its productivity and an explicit decision about whether it may be applied. No machine learning is used, and every decision the system makes can be traced to a rule or a dictionary entry.

On 499 held-out word types from news text, annotated blind and never used during development, the system reaches 71.34% exact-match accuracy against 40.48% for the published baseline, and 70.6% against 16.9% on the types that require a strip. On a 2,717-type development set it reaches 79.43% against 50.68%, with Paice's under-stemming index falling from 0.919 to 0.314.

Three observations generalise beyond this particular system. Exact-match accuracy is not a sufficient measure for stemming in a language where a large share of word types carry no affix, because it rewards inactivity; reporting the two halves of the task separately costs nothing and prevents the error. An evaluation set drawn from a single orthographic convention will conceal defects that a second source exposes immediately, as the ye variants did here. And when a rule-based system finds no affix, the useful response is to ask what the inventory is missing rather than to remove characters regardless: the one rule in this work that guessed a boundary instead of finding one was measured against three references, lost on both careful annotations, and was removed.

The stemmer, the affix inventory with its sources, the verb dictionary, both annotated sets and the scripts that

reproduce every figure in this paper are released under an MIT licence.

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APPENDIX A: THE COMPLETE AFFIX INVENTORY

The 117 rules in full, in the order the engine holds them. "Type" gives the side and whether the rule is inflectional or derivational; "min" is the shortest stem the rule is allowed to leave; "strip" is whether the engine may remove the affix at all. The entries marked no are documented and never applied, for the reasons given in Section IV.B. This table is generated directly from the rule objects by `paper/make_inventory_tables.py`, so it cannot drift from the implementation.

TABLE A.I. THE COMPLETE AFFIX INVENTORY

Affix	Type	min	POS	Productivity	Confidence	Strip
Nominal inflection						
تانه	suf, infl	3	N	productive	high	yes
تانو	suf, infl	3	N	productive	high	yes
گانو	suf, infl	2	N	productive	high	yes
گانې	suf, infl	2	N	productive	high	yes
کونه	suf, infl	2	N	productive	high	yes
گونو	suf, infl	2	N	productive	high	yes
اتو	suf, infl	3	N	productive	high	yes
انو	suf, infl	2	N	productive	high	yes
انې	suf, infl	3	N	productive	high	yes
جات	suf, infl	3	N	productive	high	yes
ونه	suf, infl	2	N	productive	high	yes
ونو	suf, infl	2	N	productive	high	yes
کان	suf, infl	2	N	productive	high	yes
ات	suf, infl	3	N	productive	high	yes
ان	suf, infl	3	N	productive	high	yes
ون	suf, infl	3	N	productive	high	yes
وو	suf, infl	3	N	productive	high	yes
وې	suf, infl	3	N	productive	high	yes
يو	suf, infl	3	N	productive	high	yes
ه	suf, infl	3	N/ADJ	productive	high	yes
و	suf, infl	3	N/ADJ	productive	high	yes
ى	suf, infl	3	N/ADJ	productive	high	yes
ې	suf, infl	2	N/ADJ	productive	high	yes
Verbal inflection						
ېدا	suf, infl	2	V	productive	high	yes
ېدل	suf, infl	2	V	productive	high	yes
ېده	suf, infl	2	V	productive	high	yes
ېدو	suf, infl	2	V	productive	high	yes
ېږم	suf, infl	2	V	productive	high	yes
ېږو	suf, infl	2	V	productive	high	yes
ېږى	suf, infl	2	V	productive	high	yes

لو	suf, infl	2	V	productive	high	yes
ول	suf, infl	2	V	productive	high	yes
ہد	suf, infl	2	V	productive	high	yes
ہز	suf, infl	2	V	productive	high	yes
ئ	suf, infl	3	V	productive	high	yes
ل	suf, infl	2	V	productive	low	no
Derivation: nouns						
یدونکی	suf, der	3	N/ADJ	productive	high	yes
ہدونکی	suf, der	3	N/ADJ	productive	high	yes
خانہ	suf, der	3	N	productive	high	yes
دانی	suf, der	3	N	borrowed	medium	yes
ستان	suf, der	3	N	borrowed	medium	no
غالی	suf, der	3	N	lexicalized	low	no
والو	suf, der	3	N	productive	high	yes
والی	suf, der	3	N	productive	high	yes
ونکو	suf, der	3	N	productive	high	yes
ونکی	suf, der	3	N/ADJ	productive	high	yes
ونکپ	suf, der	3	N/ADJ	productive	high	yes
کوئی	suf, der	3	N	restricted	low	no
گلوی	suf, der	3	N	restricted	high	yes
گوئی	suf, der	3	N	restricted	low	no
اوی	suf, der	3	N	restricted	medium	no
بان	suf, der	3	N	borrowed	high	yes
توب	suf, der	3	N	productive	high	yes
تون	suf, der	3	N	productive	high	yes
تیا	suf, der	2	N	productive	high	yes
خور	suf, der	3	N	borrowed	medium	yes
دار	suf, der	3	N	productive	high	yes
دان	suf, der	4	N	borrowed	medium	yes
زار	suf, der	3	N	borrowed	medium	yes
لیک	suf, der	3	N	productive	high	yes
نکو	suf, der	3	N/ADJ	productive	high	yes
نکی	suf, der	3	N/ADJ	productive	high	yes

نکې	suf, der	3	N/ADJ	productive	high	yes
وال	suf, der	3	N	restricted	medium	no
وان	suf, der	3	N	restricted	low	no
ولي	suf, der	3	N	restricted	low	no
وتی	suf, der	3	N	productive	high	yes
وکی	suf, der	3	N	lexicalized	low	no
پال	suf, der	3	N/ADJ	productive	high	yes
پوه	suf, der	3	N	productive	high	yes
ځای	suf, der	3	N	productive	high	yes
ځيو	suf, der	3	N	productive	high	yes
ښنه	suf, der	3	N	restricted	high	yes
کار	suf, der	3	N	productive	high	yes
گار	suf, der	3	N	borrowed	high	yes
گره	suf, der	3	N	productive	high	yes
گری	suf, der	3	N	productive	high	yes
يال	suf, der	3	N	restricted	medium	no
ار	suf, der	4	N	restricted	medium	no
اکي	suf, der	4	N	restricted	high	yes
نه	suf, der	3	N	productive	high	yes
نې	suf, der	3	N	productive	medium	no
ځی	suf, der	3	N	productive	high	yes
چي	suf, der	3	N	lexicalized	low	no
ښت	suf, der	3	N	productive	high	yes
گر	suf, der	3	N	productive	high	yes
کی	suf, der	3	N	productive	high	yes
یت	suf, der	3	N	productive	high	yes
ا	suf, der	4	N	restricted	low	no
ت	suf, der	4	N	restricted	medium	no
ک	suf, der	4	N	restricted	medium	no
ی	suf, der	3	N/ADJ	productive	high	yes
Derivation: adjectives						
انه	suf, der	3	ADJ	productive	high	yes
مند	suf, der	3	ADJ	productive	high	yes

ناک	suf, der	3	ADJ	productive	high	yes
چکی	suf, der	3	ADJ	productive	high	yes
یزہ	suf, der	3	ADJ	productive	high	yes
یزو	suf, der	3	ADJ	productive	high	yes
جن	suf, der	3	ADJ	restricted	high	yes
من	suf, der	3	ADJ	productive	high	yes
نی	suf, der	3	ADJ	productive	medium	no
نی	suf, der	3	ADJ	productive	medium	no
ور	suf, der	3	ADJ	restricted	medium	no
ور	suf, der	3	ADJ	productive	high	yes
ین	suf, der	3	ADJ	restricted	medium	no
یز	suf, der	3	ADJ	productive	high	yes
Derivation: verbs						
لپ	suf, der	3	ADJ/V	productive	high	yes
Prefixes						
بلا	pre, der	3	ADJ	borrowed	medium	yes
بیا	pre, der	3	V/N	productive	high	yes
غیر	pre, der	3	ADJ	borrowed	high	no
نیم	pre, der	3	ADJ/N	restricted	high	yes
بپ	pre, der	3	ADJ/N	productive	high	no
سر	pre, der	4	N	restricted	medium	no
ضد	pre, der	3	N	borrowed	medium	yes
لا	pre, der	3	—	borrowed	medium	yes
نا	pre, der	3	ADJ/N	productive	high	no
هم	pre, der	3	—	productive	high	yes